PETE Process: Photon-Enhanced Thermionic Emission

Based on the paper by Jared W. Schwede et al.

Nature Materials, 2010

Presentation based on the paper: Photon-enhanced thermionic emission for solar concentrator systems

The PETE Process and Its Solutions

What is the PETE Process?

Photon-Enhanced Thermionic Emission (PETE) combines quantum and thermal mechanisms into a single physical process:

Electrons are excited by solar radiation to the conduction band

Electrons thermalize and diffuse in the cathode

Electrons with sufficient energy are emitted into vacuum and collected at the anode

Problems Solved by PETE:

Thermalization losses in photovoltaic cells (about 50% of energy)

Sub-bandgap photon absorption losses

Low efficiency of photovoltaic cells at high temperatures

Low efficiency of thermal engines at low temperatures

$$J_C = AT_C^2 \exp(-(\varphi-(E_{F,n}-E_F))/kT_C)$$

PETE harnesses both the high energy of photons and available thermal energy

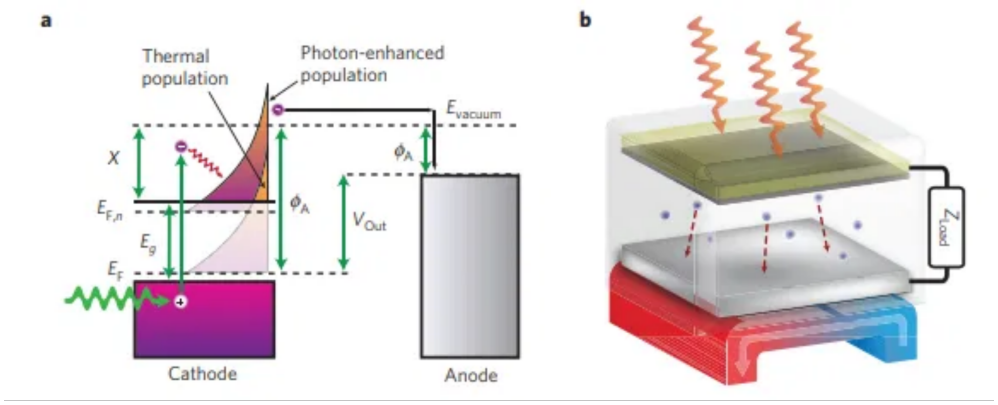


Figure 1: (a) Energy diagram of the PETE process (b) Possible implementation of a PETE converter

Three Regimes of Electron Emission

Three Regimes of Electron Emission:

Photoemission Regime:
At low temperatures, electrons are emitted directly from photonic excitation without significant thermalization.

PETE Regime:
At intermediate temperatures, photonically excited electrons undergo thermalization and are emitted with the help of thermal energy.

Thermionic Regime:
At very high temperatures, pure thermionic emission dominates regardless of photonic excitation.

Characteristics of the Regimes:

At low temperatures, electrons in the conduction band cannot overcome the electron affinity barrier.

As temperature rises, the PETE process becomes more efficient and current increases.

At very high temperatures, pure thermionic emission dominates and current no longer depends on the number of photonically excited electrons.

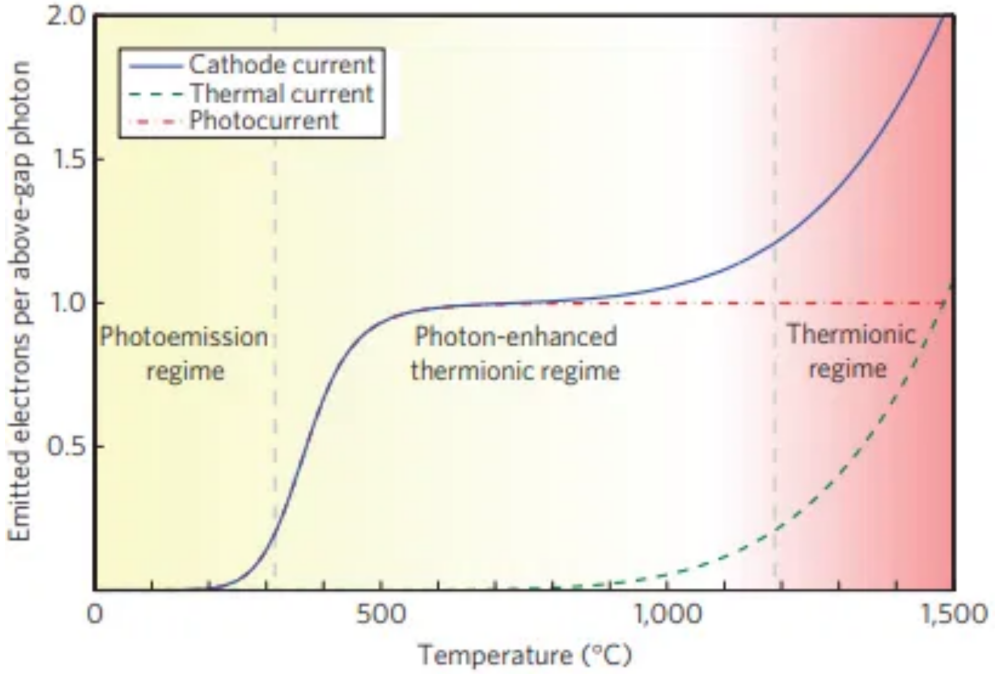


Figure 2: Three regimes of electron emission as a function of temperature

The example assumes a cathode with χ = 0.7 eV, Eg = 1.0 eV and solar concentration of ×100

Temperature-Dependent Measurements of [Cs]GaN

Experimental Setup:

Cesium-coated GaN samples were tested in an ultra-high vacuum chamber

Cesium coating reduces electron affinity to approximately 0.3-0.4 eV

Samples were illuminated with monochromatic light and electron energy distribution was measured

Experimental Results:

Energy distribution of emitted electrons shows characteristic thermal shape

Distribution width increases with temperature, confirming thermalization

At 400°C, electrons illuminated with different wavelengths (330nm and 375nm) showed identical energy distribution

Each electron excited with 3.3eV light acquired approximately 0.5eV additional thermal energy

The experiment confirms the PETE mechanism and demonstrates the "thermal boost" that enhances conversion efficiency

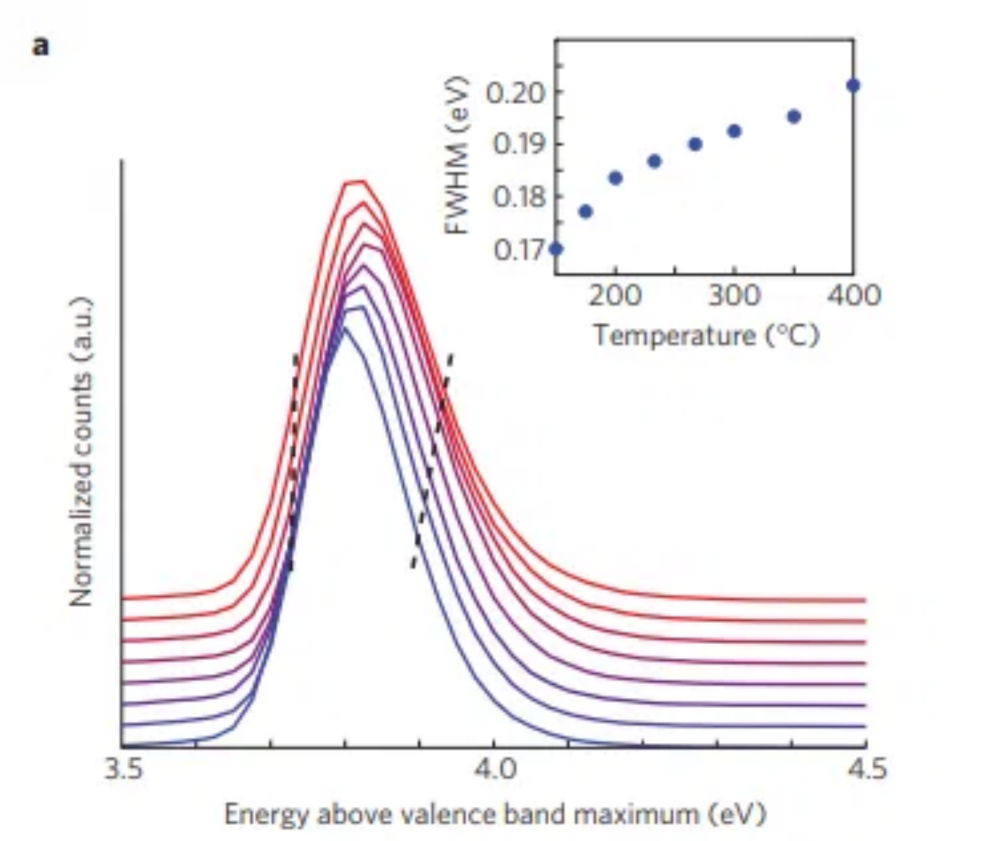


Figure 3: Energy distributions of electrons emitted from GaN at different temperatures

Theoretical PETE Efficiency

PETE Energy Conversion Efficiency:

PETE efficiency is calculated as a function of bandgap for concentrated solar spectrum

Cathode temperature and electron affinity are chosen to maximize overall efficiency

Anode temperature is set to 227°C to minimize reverse current

The chart shows output at ×1000 concentration resulting from thermal and photonic contributions

Efficiency Characteristics:

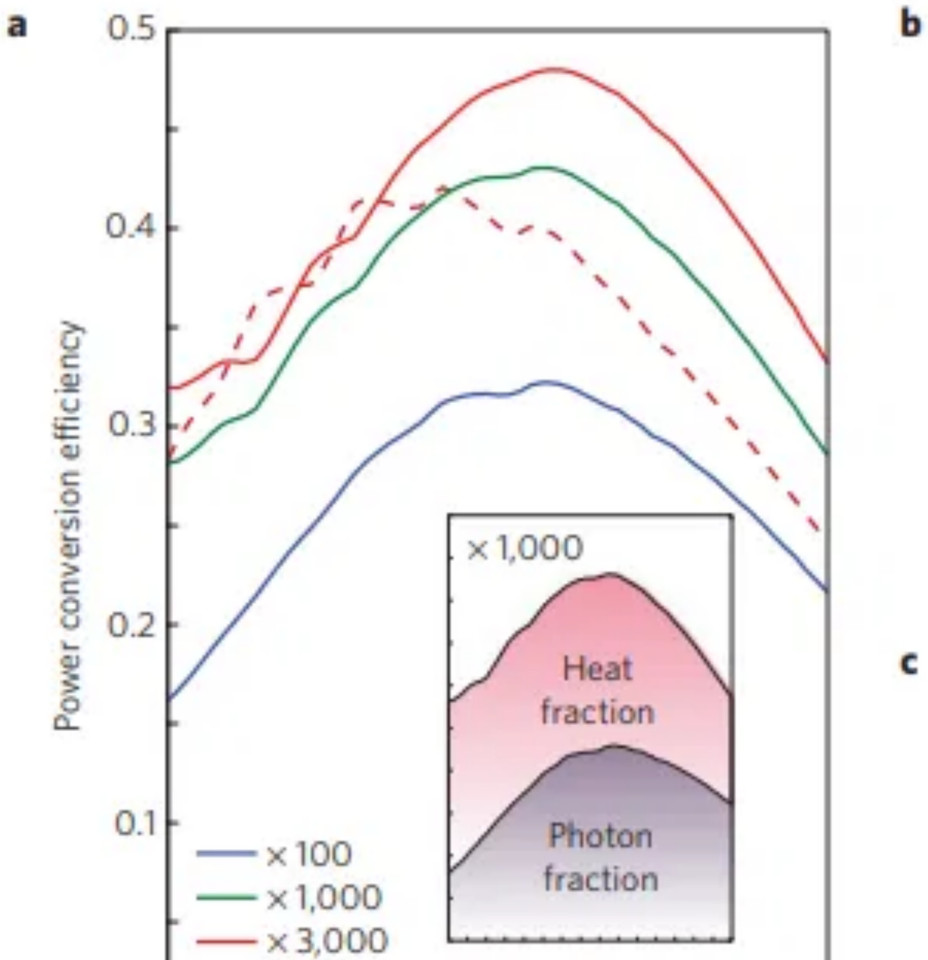
Conversion efficiency increases with higher electron affinity (χ) due to higher operating voltage

At higher temperatures, efficiency decreases almost linearly with temperature

Concentration is ×1000, and e-χ curves are shown for PETE efficiencies with identical electron affinities

Under flat-band conditions with χ = 0.4eV, temperatures at which efficiencies are maximized are shown

PETE can achieve higher conversion efficiency than conventional photovoltaic cells



Theoretical Tandem Efficiency

PETE/Thermal Tandem System:

Solar radiation is absorbed in the cathode, generating electrons and thermal radiation

The cathode emits blackbody radiation both away from the device and toward the anode

Radiative exchange between cathode and anode is minimized due to geometry

Heat not utilized in the PETE circuit is transferred to a thermal circuit, enhancing overall efficiency

Energy Conversion Efficiency of the Tandem System:

PETE/thermal tandem efficiency at ×1,000 concentration is shown in Figure 5b

The optimal temperature is 285°C with a thermo-electric engine efficiency of 31.5%

Overall conversion efficiencies exceeding 50% are possible even at low anode temperatures

The optimal bandgap is between 1.1 and 1.15eV, producing more heat than a single PETE device

The tandem system enables significantly higher conversion efficiency than either technology alone

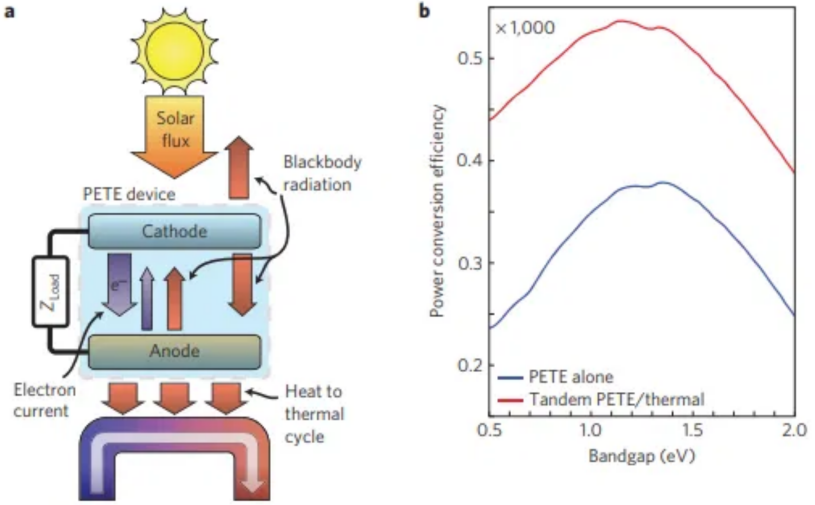


Figure 5: (a) Energy flow diagram for PETE/thermal tandem system (b) Comparison of PETE efficiency alone vs. with thermal engine

Conclusions

- The PETE process combines quantum and thermal mechanisms to create a single physical process for solar energy conversion.
- Temperature-dependent electron emission measurements from GaN provide strong evidence for the PETE mechanism.
- Theoretical efficiency of PETE devices can exceed the theoretical limits of single-junction photovoltaic cells.
- PETE converter operates at temperatures above 200°C, allowing waste heat to power a secondary thermal engine.
- A PETE/thermal tandem system can achieve overall conversion efficiency above 50%, significantly more than either technology alone.

PETE offers an innovative approach to more efficiently utilize the solar spectrum and